

SMART BUILDING DECARBONISATION IN SCOTLAND & WIDER UK

Practical frameworks and intelligent
insights for a net zero built environment

WHITE PAPER



DATA-DRIVEN

Intelligent insights for
better decisions



PRACTICAL

Actionable frameworks
for real-world estates



RESILIENT

Secure, future-ready
and risk-aware



IMPACTFUL

Accelerating progress
to net zero



Smarter Decarbonisation: Optimise. Sequence. Deliver.

A Smart Net Zero white paper helping to address the major challenges in planning and implementing commercial and public sector building decarbonisation in the UK.

Abstract

The UK's path to net zero runs through an estate that is predominantly pre-1980s and highly diverse, especially acute in Scotland's rural and island communities. Decarbonisation stalls not only on technology cost but on readiness: incomplete data, unclear sequencing, programme risk, and often-overlooked safety, security and resilience risks from increasingly smart, connected systems. Our SMART: Scotland grant-backed research translated these barriers into two newly developed practical frameworks:

1. The Building Decarbonisation Readiness Rating (BDRR)
2. The Service Maturity Action Matrix (SMAM)

Both have now been operationalised in the newly developed AI deep neural **SmartDecarb360** decision engine. Together, they convert thin data into auditable, staged plans that explicitly factor security and resilience alongside carbon, cost and comfort.

1. Why building decarbonisation is hard

Older fabric and constraints. Most of the UK's housing stock predates 1980 - around 78% of homes were built before 1980 [1]. While equivalent age statistics for non-domestic buildings are less complete, Scotland's policy research highlights significant diversity and data gaps across approximately 220,000 non-domestic buildings [4, 5]. Older fabric, legacy plant/controls, asbestos risk, heritage restrictions and limited plant space make low-temperature heat, ventilation and deep fabric upgrades complex and disruptive, especially during term times in schools or clinical operations in health settings.

System variability and delivery risk. Mixed fuel types, partial or absent BMS, and ad-hoc upgrades create site-to-site inconsistency. Scottish assets often face higher heating demand and logistics constraints (rural and island communities), short delivery windows (weather and academic calendars), and supply-chain limitations.

Capacity, governance and safety. Estates teams juggle statutory maintenance with limited analytics capacity. Safety and resilience risks from refrigerant handling, Legionella during fabric works, fire interactions, to cyber-physical risk from connected controls – all are often under-scoped during decarbonisation planning.



2. The infrastructure data problem

Many portfolios hold only high-level consumption and asset lists; telemetry is partial, vendor-siloed or inaccessible. Energy, maintenance and capital planning data live in different tools (EPCs, CAFM, BMS, spreadsheets). Raw telemetry seldom answers board-grade questions (installation complexity, disruption, wellbeing, security, resilience), which weakens investment cases and increases the chances of missing materials risks. Scotland-focused reviews emphasise inconsistent coverage and quality for non-domestic stock data [5].

3. What the research identified

From pilots and feasibility work, three repeatable insights emerged:

1. **Readiness is multi-dimensional:** Seven interacting dimensions beyond pure carbon and cost: energy impact and savings potential; installation complexity; impact against target year; resilience (cyber, climate, supply, economic); wellbeing & safety; smart/data readiness; and financial feasibility.
2. **Digital capability vs physical scope:** Digital maturity (behaviour, analytics, automation) is orthogonal to physical maturity (fabric/plant changes). Treating them separately makes 'what to do next' unambiguous at each site.
3. **Governance needs evidence and explainability:** Versioned rules, evidence lists, and uncertainty notes make decisions auditable for boards, funders and assurance partners.

4. Why these frameworks (BDRR + SMAM) and SmartDecarb360

BDRR (1–5; Baseline > Platinum). A configurable readiness rating across the seven dimensions above. Custom weightings (finance-first, climate-first, resilience-first) let organisations like councils, RSLs, NHS and education reflect policy and budget realities while keeping results comparable. A configurable readiness rating across seven dimensions. Of which Resilience is one that explicitly covers cyber-security controls, life-safety interactions (fire, smoke control, ventilation/IAQ), climate-hazard exposure, grid/backup adequacy, supply-chain and O&M resilience. Weightings adapt to organisational priorities (finance-first, climate-first, resilience-first).

SMAM (Stages 0–3). A two-axis view in Smart Maturity (behaviour > analytics > automated/predictive) vs Action Maturity (baseline planning > targeted upgrades > deep retrofit/optimisation). The matrix sequences no-regret steps for older stock, controls optimisation, ventilation and low-disruption fabric before invasive heat electrification, unless readiness indicates otherwise.

SmartDecarb360 decision engine. Combines BDRR + SMAM to triage portfolios, optimise bundles under budget/constraints, and produce board-ready packs (cost/benefit, £/tCO₂e,



disruption windows, safety/security assurance checkpoints). Starts from minimal data and improves as telemetry integrates.

5. How big is the prize? Market impact and contribution to net zero

Buildings and product use accounted for around one-fifth of UK greenhouse gas emissions in 2023. Within this, residential fuel combustion made up about 65% of buildings emissions, with commercial buildings around 17% and the public sector about 11% [2]. Non-residential buildings overall contributed roughly 5% of total UK emissions in 2023 ($\approx 20.8 \text{ MtCO}_2\text{e}$) [3]. In Scotland, buildings account for $\sim 20\%$ of national emissions, with $\sim 7\%$ from non-domestic buildings; policy aims target a 68% reduction in building emissions by 2030 and zero by 2045 [6, 7]. Readiness-led sequencing with security and resilience embedded helps accelerate early, low-disruption tonnes and protects outcomes from outages, attacks or safety incidents. This increases board confidence in organisations to commit to multi-year capital.

Translating fragmented data into readiness-led, staged plans can:

- **Accelerate near-term tonnes:** prioritise buildings with the highest feasible impact across organisations like councils, NHS boards, universities/colleges, and commercial portfolios.
- **Strengthen investment cases:** evidence-first outputs unlock finance (including blended/public funding) for heat electrification and fabric.
- **Scale governance and delivery:** standardised readouts, assurance checkpoints and repeatable board packs reduce risk in multi-year frameworks and FM/ESCO partnerships.
- **Operational resilience:** designs tolerate grid events and cyber incidents; IAQ and comfort protected during retrofits.
- **Build a data-positive flywheel:** each intervention adds metering, commissioning records and controls data, raising Smart Maturity and enabling automation over time.

6. Illustrative pathway for older commercial building stock

Our research found the approach linked with SmartDecarb360 could include:

- **Rapid assess (4–6 weeks):** Import portfolio baseline, apply weightings, compute BDRR/SMAM, record initial security posture (internet-exposed endpoints, remote access, backup), flag quick wins (controls, ventilation tuning, low-disruption fabric) vs deep packages (heat pumps + fabric + renewables) for high-readiness sites.
- **Board packs & sequencing:** Produce Tranche A (no-regret/low-disruption) and Tranche B (deep decarb) with costs, $\text{£/tCO}_2\text{e}$, disruption windows and security/safety assurance gates.
- **Pilot & calibrate (90 days):** Metered/BMS subset validates savings, refine rules, align outputs with statutory indicators, run basic cyber hardening and IAQ commissioning.
- **Scale:** Add connectors (EPC, CAFM, BMS/IoT), automate reassessments, and pursue independent assurance for Gold/Platinum ratings.



7. Conclusion

Older UK and Scottish building stock, fragmented data, and growing cyber-physical exposure make decarbonisation a sequencing and assurance challenge as much as a technical one. SmartDecarb360 addresses this by translating limited, siloed data into readiness-led, auditable action - optimising what to do first, where, and how, while embedding security and resilience into every step.

By adopting BDRR and SMAM, organisations with commercial building portfolios can accelerate near-term emissions cuts, reduce programme risk, and build toward automated, predictive operations. The result is faster, safer delivery and stronger investment cases aligned to net-zero targets.

Contact, References & Acknowledgement

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References

Key sources are listed in the References section of this paper (UK emissions statistics from DESNZ/CCC, Scotland non-domestic stock evidence from ClimateXChange/Scottish Government) Additional references and datasets are available on request.

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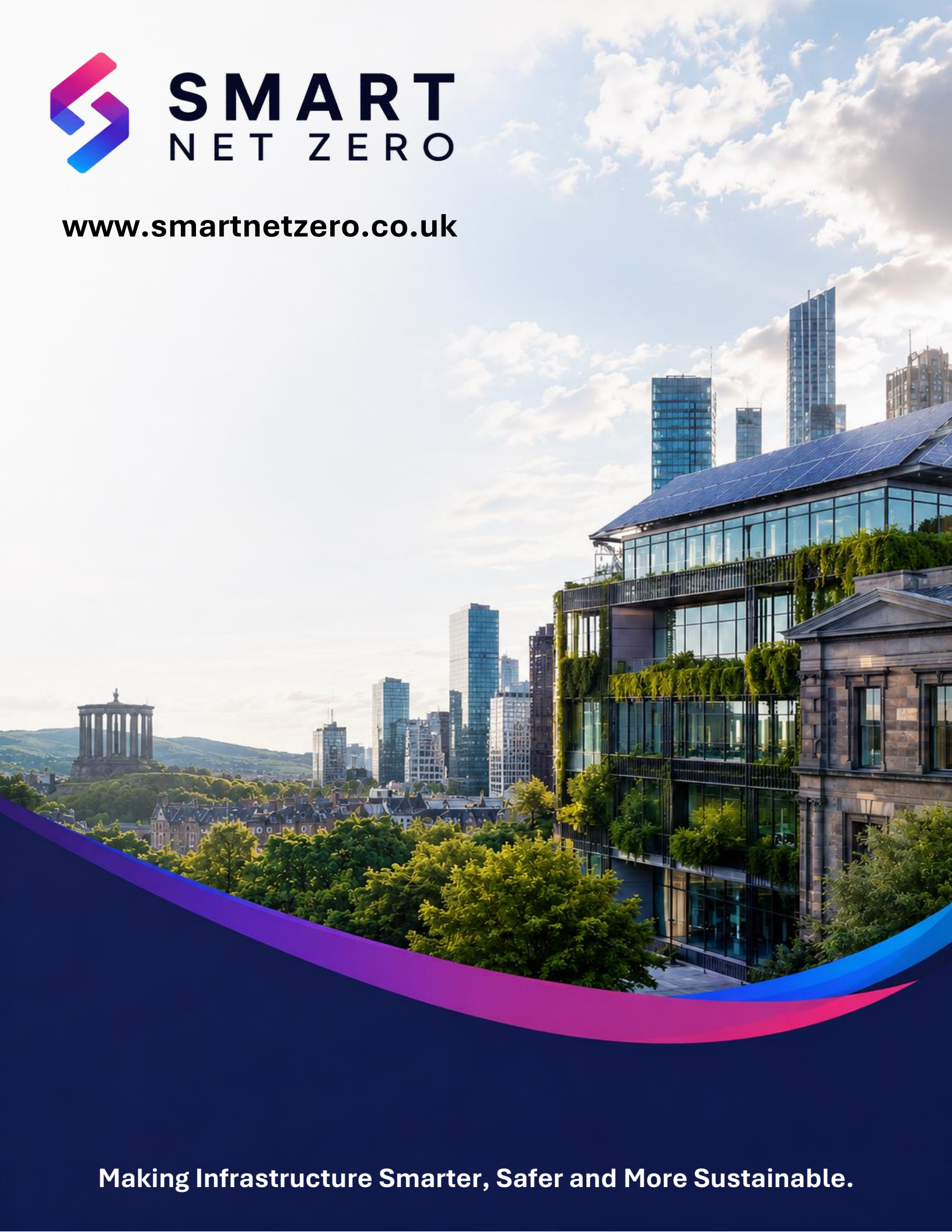


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